

SMART BIN
IoT SYSTEM

DETAILED IMPLEMENTATION PLAN

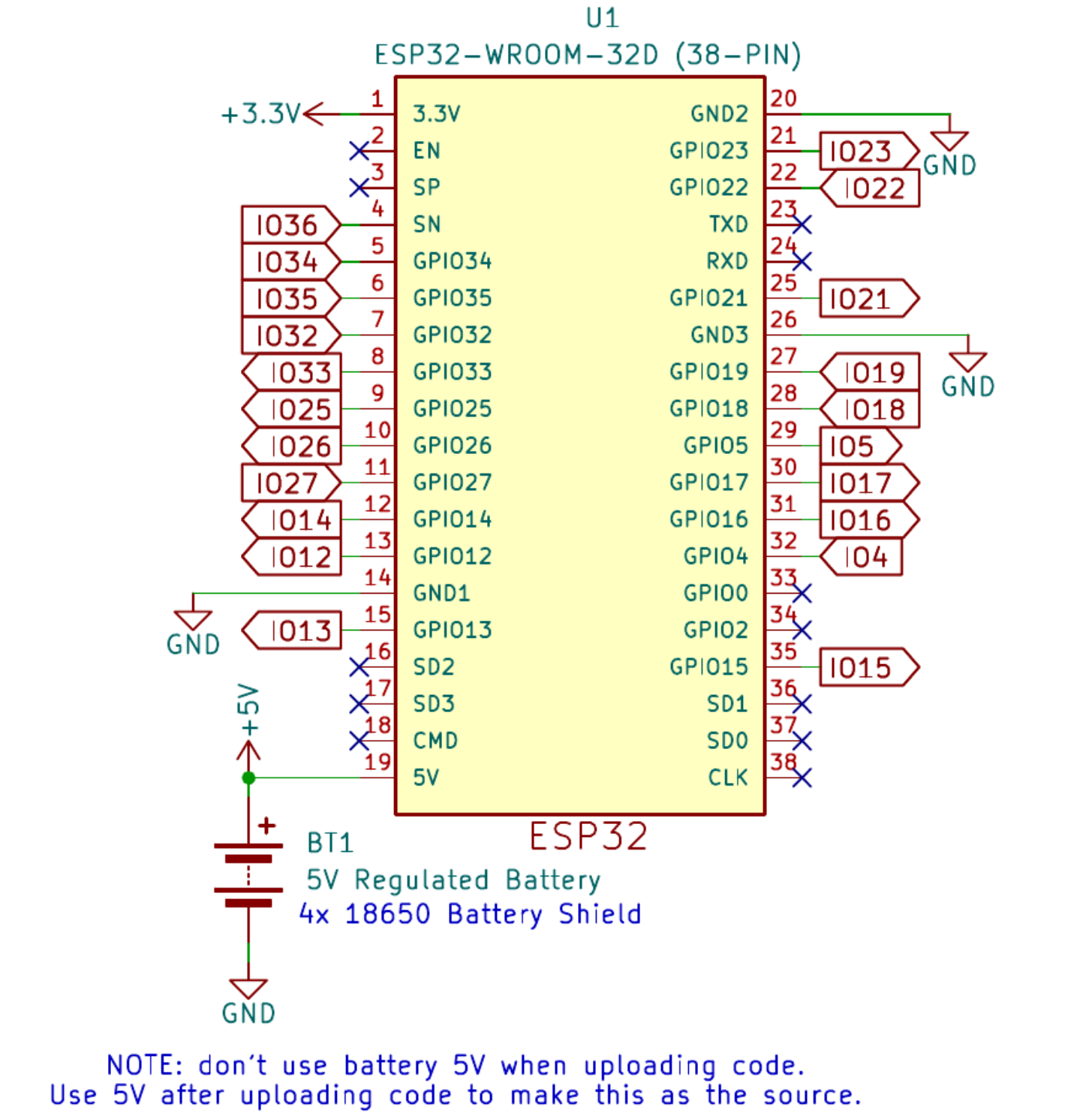
Embedded Systems | IoT Level 6
TUP Manila | Group 2 | BET-CpET 4A-M

TEAM MEMBERS

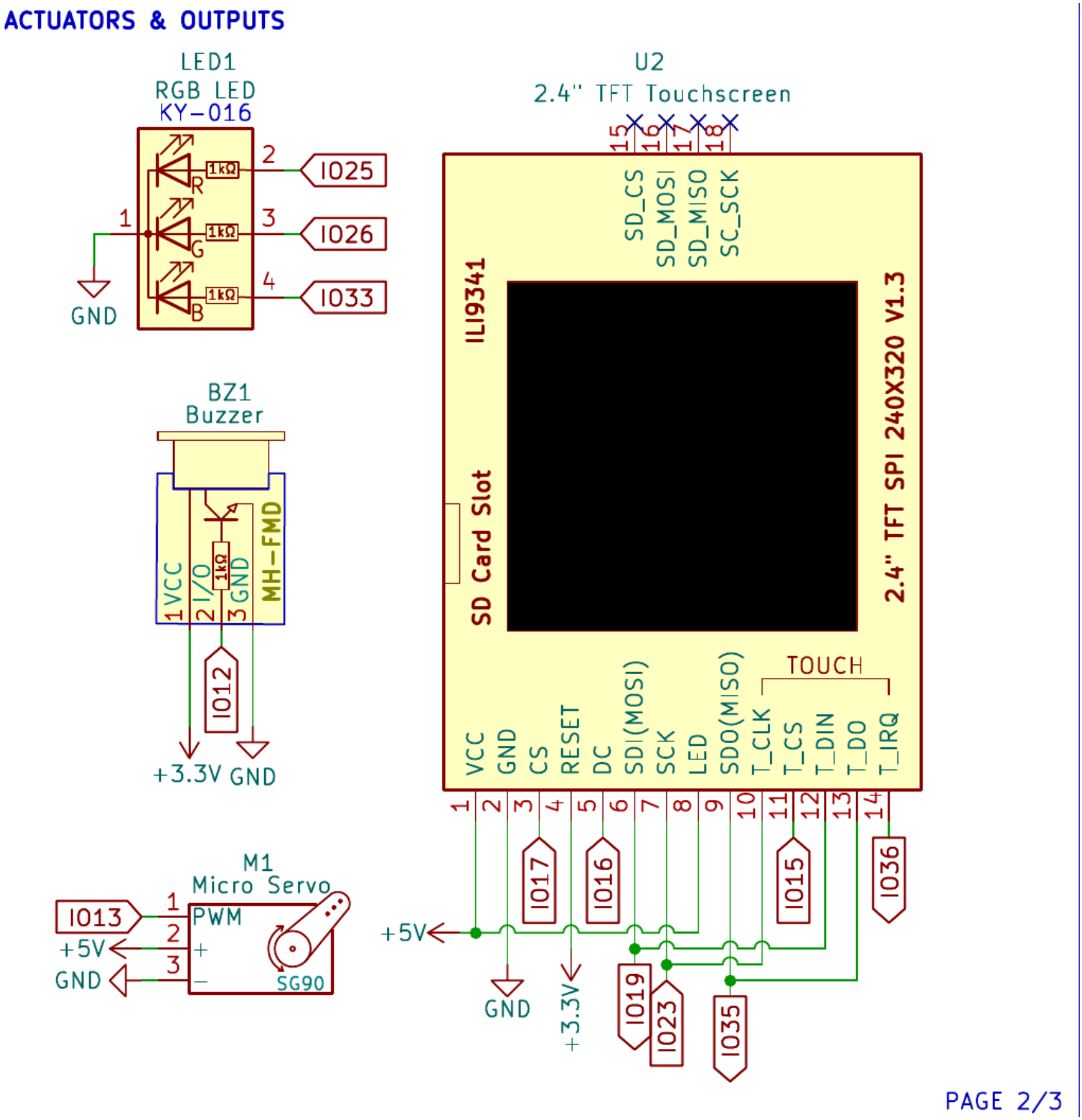
Celeste, Brian Jay - Schematic / PCB Layout
Reyes II, Genesis Walter D. - Firmware + Web Dashboard
Pepito, King Perth G. - Android Mobile App
Banogbanog, Rodel S. - Hardware Build / Physical Model
Gunayan, Jade C. - Hardware Assembly Assistant

Revised October 9, 2026 | v2 (BME680 + new schematic + Oct-Nov schedule)

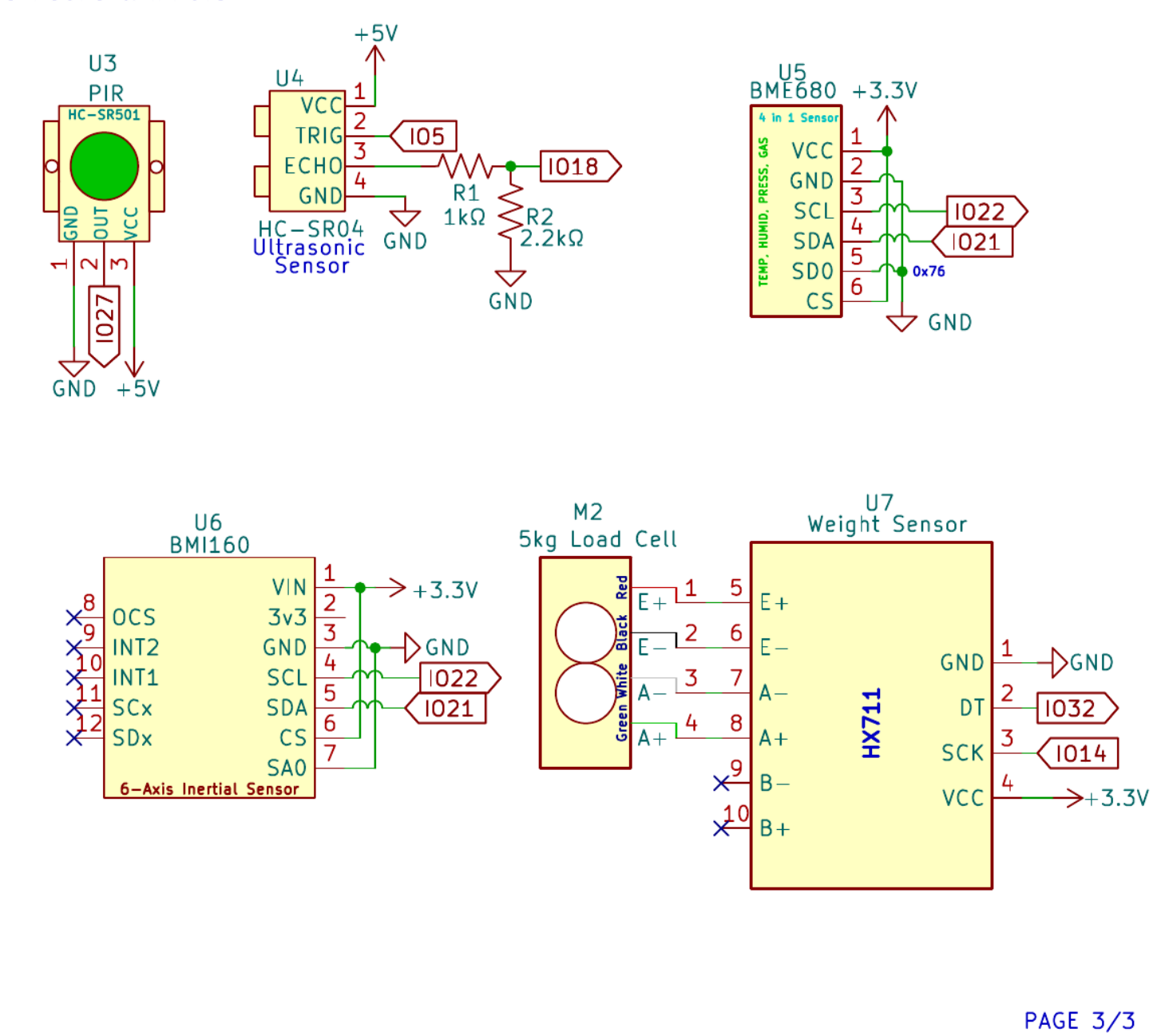
MICROCONTROLLER -- ESP32-WROOM-32D (38-pin) + 5V Battery Shield



ACTUATORS & OUTPUTS -- RGB LED, Buzzer, Servo, 2.4in TFT Touch



SENSORS & INPUTS -- PIR, HC-SR04, BME680, BMI160, HX711 + Load Cell

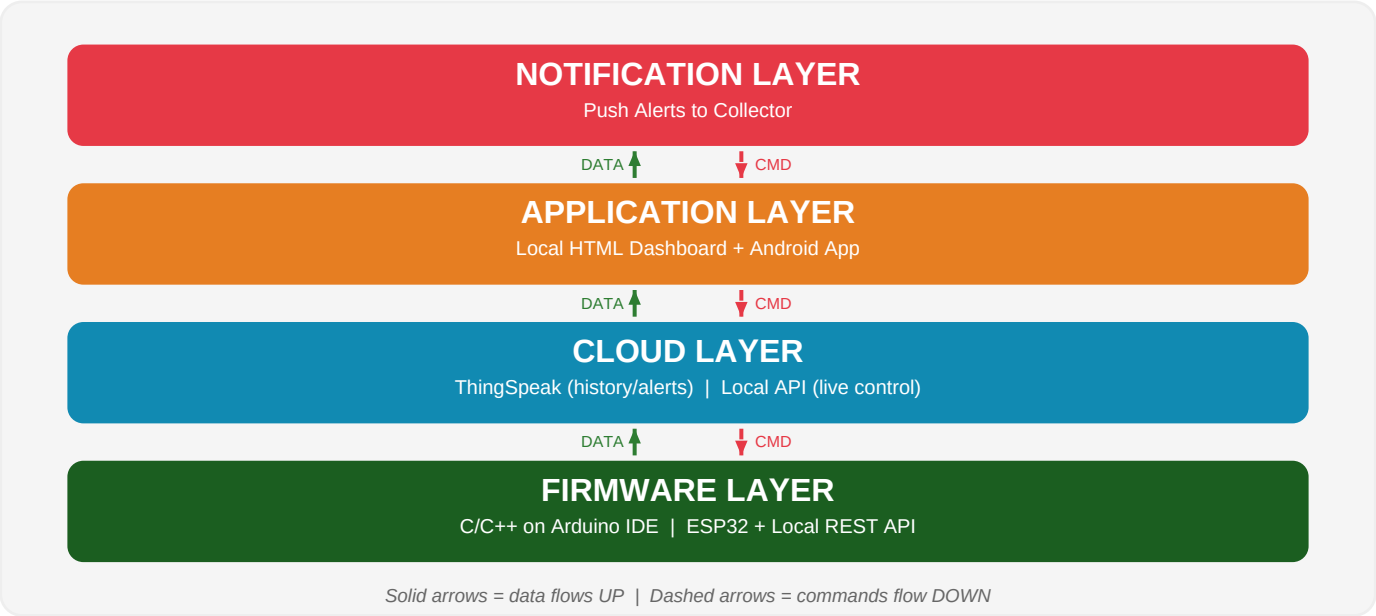


What This System Does

Two smart trash bins on a table-size model, each equipped with sensors and an ESP32 microcontroller. They measure fill level in real-time via a hybrid architecture: a local REST API on each ESP32 for fast ~1-second polling over WiFi, plus ThingSpeak cloud for historical data and alerts. Operators monitor through a local web dashboard and collectors use an Android app for alerts and collection tasks.

IoT Level 6 Architecture

Multiple independent end nodes + centralized cloud controller with bidirectional data flow.



Bidirectional Data Flow

DATA UP (Sensor to Dashboard)

- FAST: Dashboard polls ESP32 local API every ~1s
- SLOW: ESP32 posts to ThingSpeak every 15s (history)
- App uses ThingSpeak API for cloud access
- Live charts update from local API in real-time

COMMANDS DOWN (Dashboard to Bin)

- FAST: Dashboard POSTs to ESP32 local API directly
- SLOW: TalkBack queue (LOCK/COLLECT) for app
- Simulation + calibration via local API (instant)
- Firmware sets LOCK flag or enters COLLECT mode

What Runs Locally vs What Runs on ThingSpeak

LOCAL WiFi (fast, ~1s)

- All live sensor readings (direct ESP32 polling)
- Simulation toggles (Auto/Manual per sensor)
- Calibration settings changes (instant apply)
- LOCK and COLLECT commands (direct to firmware)
- Dashboard UI (local .html file in browser)
- Touchscreen display (on-device, no network)

ThingSpeak Cloud (background, 15s)

- Historical data storage (graphs over time)
- React alerts (server-side threshold checks)
- ThingHTTP webhooks (push notifications)
- Mobile app data when off local WiFi
- Channel 3: calibration backup (synced)
- Channel 4: simulation state backup (synced)

DURING DEMO: Everything the panelists see in real-time is local WiFi. ThingSpeak quietly logs in the background.

Components Per Bin (Quantity x2 for full system)

ESP32 — Microcontroller

- Built-in Wi-Fi (802.11 b/g/n)
- Dual-core 240 MHz processor
- 34 GPIO pins for all peripherals
- Central hub: reads sensors, controls actuators, sends data

HC-SR04 — Ultrasonic Sensor

- Measures distance from lid to trash surface
- Range: 2 cm to 400 cm
- Mounted at inner top of bin, pointing downward
- 5 readings taken, MEDIAN used (filters noise)

PIR — Motion Sensor

- Detects hand approach within ~3 meters
- Triggers servo to open lid automatically
- Mounted on outer front face of bin
- Digital output: HIGH when motion detected

SG90 — Servo Motor

- Controls lid: OPEN / CLOSE positions
- 0 deg = closed, 90 deg = open (angles configurable via dashboard)
- Mounted flat (horizontal) on bin top surface, arm pushes lid open
- LOCK = software state (servo stays closed, firmware refuses open)

RGB LED — Status Indicator

- GREEN: fill level below 50%
- AMBER: fill level 50% to 79%
- RED: fill level 80% and above (threshold)
- Visible from outside, quick visual check

Buzzer — Passive Buzzer

- Passive type: frequency-controlled for distinct tones
- Lid open beep (~1kHz, short) | Tamper alarm (2kHz/500Hz alternati...
- Threshold warning (~1.5kHz, 2s) when bin full/temp/gas spike
- Mounted near PIR sensor on front face, silenced during COLLECT

TFT — 2.4" Touch Display

- SPI display (ILI9341, 240x320) + XPT2046 resistive touch, already ...
- Shows fill %, temp, humidity, weight, air quality (IAQ) + dual status
- COLLECT requires PIN pad entry. TAMPER screen shows live cou...
- Runs on SPI, so the I2C bus is left for BMI160 + BME680

5V PSU — Power Supply

- Bin A: 4x18650 Shield V9 (5V/3A+3V/1A) | Bin B: 2x18650 Shield ...
- Stored in SEALED compartment BELOW divider wall
- Completely separated from trash chamber
- Powers ESP32 and all peripherals; rechargeable via Micro USB

BME680 — Temp / Hum / Pressure

- Replaces DHT22 + MQ-135: one I2C module, 3.3V, addr 0x76 (SD...
- Temperature (-40 to 85 C), humidity (0-100%RH), pressure (300-11...
- Offsets (temp_off, hum_off, press_off) set from dashboard calibration
- Air port hole must face bin air, away from ESP32/regulator heat

BME680 — Gas / Air Quality (IAQ)

- Gas resistance falls when VOCs from rotting waste rise
- Reported as IAQ index 0-500 (lower = cleaner) + accuracy 0-3
- Bosch BSEC library preferred; else rolling-baseline index
- Warm-up minutes after power-on; first-use burn-in hours to days

HX711 — Load Cell (5kg)

- Measures actual trash weight in grams
- 5kg capacity, mounts at bottom of trash chamber
- HX711 amplifier board converts strain to digital
- Validates ultrasonic: low weight + high fill = air gaps

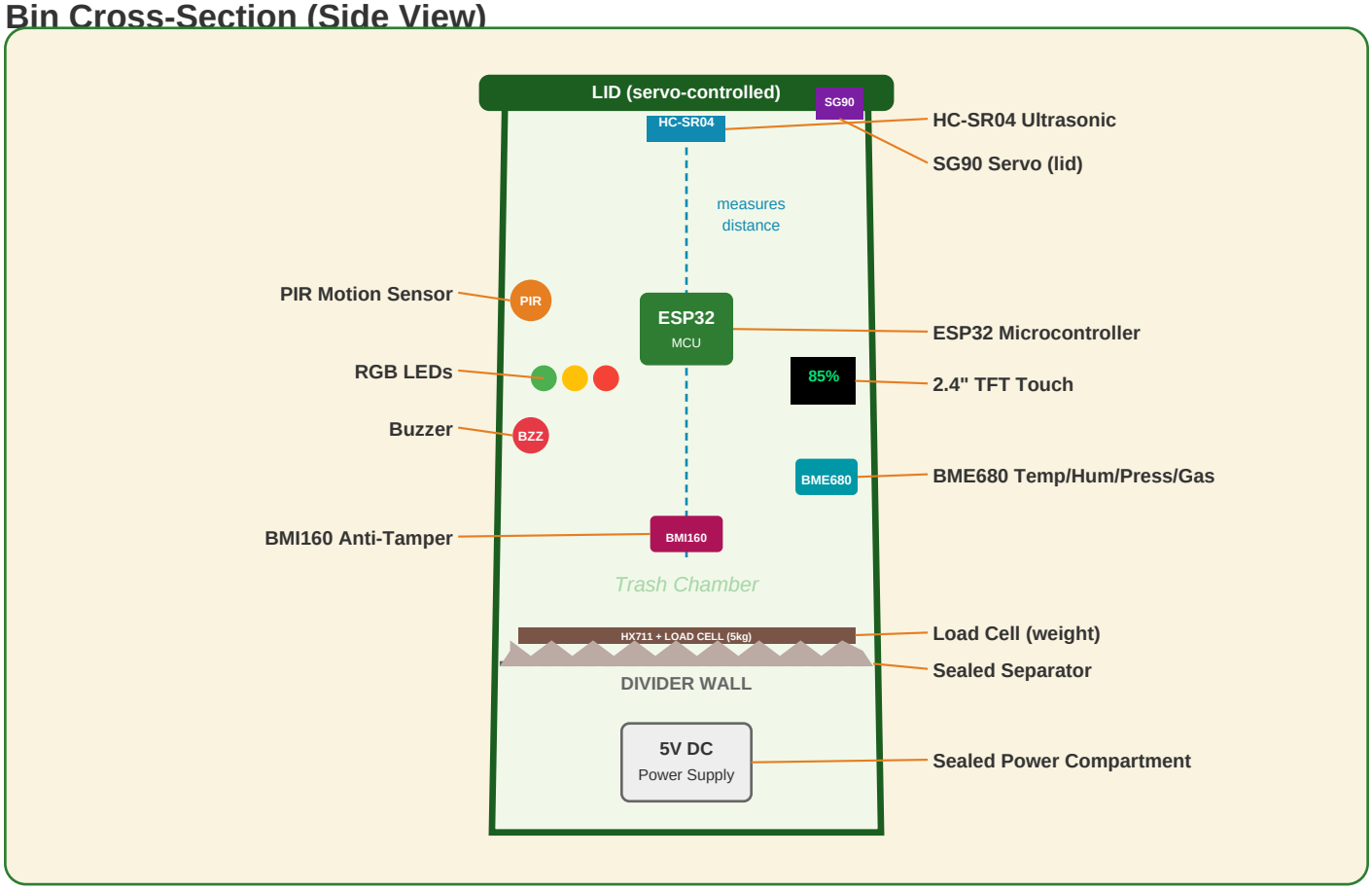
BMI160 — Anti-Tamper

- 6-axis accel + gyro (I2C bus, drop-in for MPU6050)
- Detects tilt/lift: tamper alert if angle > threshold
- Shares I2C with BME680: BMI160 at 0x68 (SA0=GND), BME680 at...
- Threshold adjustable from dashboard (Channel 3)

BIN PHYSICAL LAYOUT

Cross-Section View & Wiring

Bin Cross-Section (Side View)



Power supply is physically isolated from trash. All electronics are above the divider wall except the PSU.

Physical Build Notes

- Table-size model: approximately 30-40 cm tall per bin, two bins side by side
- Material: acrylic, wood, or 3D-printed enclosure (transparent front recommended for demo)
- All sensor wires routed internally, no external dangling cables
- PCB (if feasible) replaces breadboard for cleaner internal layout

Sensor Reading & Calibration

The ultrasonic sensor takes 5 distance readings. The MEDIAN value is used to eliminate noise spikes.

That median distance is then converted to a fill percentage using this calibration formula:

CALIBRATION FORMULA

$$\text{Fill \%} = \frac{\text{D_EMPTY} - d}{\text{D_EMPTY} - \text{D_FULL}} \times 100$$

WHERE:

- **d** = current median distance reading from HC-SR04 (in cm)
- **D_EMPTY** = user-configurable in dashboard Settings (Channel 3), default 30 cm -- no reflash needed
- **D_FULL** = user-configurable in dashboard Settings (Channel 3), default 5 cm -- no reflash needed

Main Loop (runs continuously on each ESP32)

1. **Read HC-SR04**
   Take 5 distance measurements, compute median
2. **Calculate Fill %**
   Apply calibration formula above
3. **Read BME680**
   Temperature, humidity, pressure over I2C (0x76); apply offsets from calibration
4. **Compute IAQ**
   BME680 gas resistance -> IAQ 0-500 + accuracy (BSEC or rolling baseline)
5. **Read Weight**
   HX711 load cell reading in grams (0-5000g)
6. **Update LEDs**
   Green (<50%) | Amber (50-79%) | Red (80%+)
7. **Update TFT Touch**
   Display fill %, temp, humidity, weight, IAQ on 2.4" screen. Lid: OPEN/CLOSED. Bin: OK/LOCKED/TAMPER/COLLECTING
8. **Check Threshold**
   Fill >= 80% OR temp/gas spike: warning tone (1.5kHz, 2s), lock if auto-lock
9. **Send to ThingSpeak**
   HTTP POST all 8 fields to cloud every 15s (history/alerts path)
10. **Check Tamper**
    Tilt > threshold: alarm + 10s countdown. Stable in 10s = auto-clear. 15s = locked (manual clear). SKIP in COLLECT
11. **Check TalkBack**
    Poll for LOCK or COLLECT commands (cloud path)
12. **Apply Sim Bitmask**
    Skip writing fields set to Manual (operator's value via local API stays)
13. **Check Touch**
    COLLECT tapped: PIN pad shown, correct PIN starts collect mode. DONE ends it (no PIN)
14. **Check PIR**
    Motion AND bin OK (not locked/tamper/collecting): beep + open lid, hold while PIR HIGH, 5s min before close

TIMING (HYBRID): Local REST API serves data on demand (~1s polling from dashboard). PIR is interrupt-based. ThingSpeak POST every 15s (cloud/history only). ESPAsyncWebServer handles local + cloud in parallel.

LOCAL REST API (ESPAsyncWebServer on each ESP32)

```
GET  /api/status      -- returns all sensor data as JSON (dashboard polls this every ~1s)
POST /api/simulate     -- set sim bitmask + manual values (instant, no 15s wait)
POST /api/calibrate    -- update calibration settings (instant, stored in EEPROM)
POST /api/command      -- LOCK / COLLECT (direct to firmware, no TalkBack delay)
```

SIMULATION MODE (per-sensor, fine control -- see page 16)

- Each sensor has its own Auto/Manual flag (bitmask stored in EEPROM), not one global switch
- Manual sensors: ESP32 stops overwriting that field; operator's value via local API stays live
- Auto sensors: ESP32 keeps reading real hardware and writing normally, same loop as always
- Operator toggles via local API (POST /api/simulate) for instant response, synced to Channel 4
- Both dashboard and app show a SIMULATION MODE banner whenever any sensor is in Manual
- Bitmask stored in EEPROM so it survives reboot. Real sensors resume instantly when set back to Auto

HYBRID ARCHITECTURE: FAST PATH = local ESP32 REST API (ESPAsyncWebServer, ~1s polling). SLOW PATH = ThingSpeak (15s, history). Dashboard uses fast path for live control. Mobile app can use either. Both run in parallel on each ESP32.

Why ThingSpeak (Not AWS, Not Thingsboard)

ThingSpeak	AWS IoT	Thingsboard
Free tier: 4 channels	Pay-per-use (costs money)	Free but self-hosted
Simple REST API (HTTP)	Needs X.509 certificates	Complex MQTT setup
Built-in alerts (React)	Must configure SNS/Lambda	Custom rule chains
TalkBack command queue	IoT Shadow + Rules Engine	RPC commands
15-min setup	Hours of IAM/cert config	Server deployment needed
Perfect for 2 bins	Built for 10,000+ devices	Overkill for demo

ThingSpeak Channel Setup

Bin A (Channel 1)

- Field 1: fill_percentage (0-100)
- Field 2: distance_cm (raw reading)
- Field 3: lid_state (0=closed, 1=open)
- Field 4: bin_status (0=OK, 1=locked, 2=tamper, 3=collecting)
- Field 5: temperature (C, BME680, offset applied)
- Field 6: humidity (%RH, BME680)
- Field 7: iaq (0-500 index, BME680 gas)
- Field 8: weight_grams (0-5000g, HX711)

Bin B (Channel 2)

- Field 1: fill_percentage (0-100)
- Field 2: distance_cm (raw reading)
- Field 3: lid_state (0=closed, 1=open)
- Field 4: bin_status (0=OK, 1=locked, 2=tamper, 3=collecting)
- Field 5: temperature (C, BME680, offset applied)
- Field 6: humidity (%RH, BME680)
- Field 7: iaq (0-500 index, BME680 gas)
- Field 8: weight_grams (0-5000g, HX711)

Calibration Settings
(Channel 3) -- Operator writes, ESP32 reads

- F1: distance_empty_cm (30) -- HC-SR04 reading when bin empty
- F2: distance_full_cm (5) -- HC-SR04 reading when bin full
- F3: fill_threshold (80%) -- fill level that triggers alert
- F4: temp_threshold (40C) -- temperature spike limit
- F5: gas_threshold (200 IAQ) -- BME680 air-quality limit
- F6: tilt_threshold (30deg) -- BMI160 tamper angle
- F7: weight_cap_g (4500) -- HX711 near-full weight
- F8: auto_lock_enabled (0/1) -- auto-lock bin when full
- EEPROM: servo_closed/servo_open (0/90) -- servo angles
- EEPROM: collect_pin (1234) -- 4-digit PIN for COLLECT

Simulation Control
(Channel 4) -- Operator writes, ESP32 reads

- F1: sim_mask_bin_a -- bitmask, 1 bit per sensor set to Manual
- F2: sim_mask_bin_b -- bitmask, 1 bit per sensor set to Manual
- F3: sim_active_a (0/1) -- any bit set = SIMULATION banner shows
- F4: sim_active_b (0/1) -- any bit set = SIMULATION banner shows
- Bits 0-6: fill, weight, gas, temp/hum, tamper, motion, lid
- Manual values are written directly into Ch1/Ch2 fields --
- ESP32 just skips overwriting whichever field is flagged Manual
- Fields 5-8 reserved / spare

ThingSpeak Built-In Services

React — Threshold Alert Engine

- Rules: fill >= 80%, OR temp spike, OR gas above threshold triggers alert
- One React app per channel (so 2 total, one for Bin A, one for Bin B)
- Runs server-side on ThingSpeak, no code on ESP32 needed for alerting
- React is ThingSpeak's alert system name, NOT the JavaScript framework

TalkBack — Command Queue (Cloud Path)

- Each bin has its own TalkBack queue
- Commands: LOCK, COLLECT (simulation handled via local API, not TalkBack)
- Fallback for mobile app when not on same WiFi as ESP32s
- Dashboard uses local API (POST /api/command) instead -- instant, no 15s wait

ThingHTTP — Webhook Sender

- Triggered by React when threshold is crossed
- Sends HTTP request to your notification endpoint
- For this project: triggers Android app push notification
- Can also be used to log alerts to a Google Sheet (bonus)

Dashboard Overview

- **Built with:** HTML + CSS + JavaScript + Chart.js (single .html file, no backend server)
- **Runs from:** Local .html file opened in browser (file:// or double-click). No hosting needed.
- **Live data:** Polls ESP32 local REST API every ~1s over WiFi (GET /api/status)
- **History:** Fetches ThingSpeak API for charts and historical data
- **Access:** Open the .html file on any laptop on the same WiFi network as the ESP32s

Role-Based Views (G1 Feedback)

On page load, user selects their role. This shows/hides HTML elements. Same data, different views.

ADMIN / OPERATOR

- Live fill-level charts (Chart.js, updated every ~1s from local API)
- Both bins displayed side by side
- Fill history over time (from ThingSpeak)
- Collection Priority panel (sorted by fill %)
- LOCK button per bin (POST /api/command on local API, instant)
- Settings panel: distance empty/full, all thresholds (POST /api/calibrate)
- Simulation panel: per-sensor Auto/Manual toggle (POST /api/simulate)
- Estimated time-to-full per bin
- All sensor data visible (dist, PIR, temp, gas, weight, tamper)

COLLECTOR

- Alert banner (which bins need collection)
- Bin labels / location identifiers
- Collect button (marks bin as emptied)
- Simple, minimal interface
- No charts or history (not needed for this role)
- No LOCK button (only admin can lock)
- No settings or simulation controls (view-only)
- SIMULATION MODE banner shown if any sensor is Manual
- Sound/visual notification on new alert

Collection Priority Panel (Professor's Feedback)

The dashboard sorts bins by fill percentage, highest first. This is the route optimization the professor asked about. At table-size with 2 bins, 'routing' means: go to the fullest bin first.

COLLECTION PRIORITY

1. **Bin A** - 92% full **URGENT**
2. **Bin B** - 45% full **NORMAL**

With more bins, this list grows and becomes real route optimization. Algorithm: sort descending by fill %.

Estimated Time-to-Full (G1 Calendar Feedback)

If the dashboard has two data points over time, it can estimate when a bin will reach 80%:

TIME-TO-FULL ESTIMATE

$$T = \frac{80 - \text{current_fill}}{\text{fill_rate_per_hour}}$$ hours until 80%

App Overview

- **Built with:** Android Studio (Java or Kotlin)
- **Purpose:** Collector's personal device for alerts and bin collection tasks
- **Data source:** ThingSpeak REST API (cloud path) -- or local ESP32 API if on same WiFi
- **Notifications:** Firebase Cloud Messaging (FCM) for push alerts, triggered by ThingHTTP

App Screens

Home / Dashboard

- Bin cards with fill %, status color, and dual status
- Lid: OPEN/CLOSED + Bin: OK/LOCKED/TAMPER/COLLECTING
- Green/Amber/Red indicator matching bin LEDs
- Tap a bin to open Bin Detail with full sensor data
- Pull-to-refresh to update data
- SIMULATION MODE banner at top if any sensor is Manual

Bin Detail

- All sensor readings: temp, humidity, weight, gas, fill %
- Dual status indicators (same as touchscreen)
- Estimated time-to-full (hours until 80%, from fill rate)
- COLLECT button (writes COLLECT to TalkBack queue)
- History/trend if available from ThingSpeak

Alerts

- List of recent threshold alerts
- Shows which bin, what fill %, and when it triggered
- Push notification appears even when app is closed
- Tap notification to open directly to that bin's detail

Settings

- Threshold values shown read-only (collector cannot edit)
- Collect PIN visible (read-only, PIN entry is touchscreen-only)
- Notification preferences (sound on/off)
- ThingSpeak API key configuration

Push Notification Flow



SIMPLER ALTERNATIVE (if FCM setup is too complex):
The app polls ThingSpeak every 15 seconds. If fill >= 80%, show a local notification. No FCM needed.

API Endpoints

Local ESP32 REST API (fast path, same WiFi):

Live data: `GET http://192.168.1.101/api/status (Bin A) | .102 (Bin B)`

Send command: `POST http://192.168.1.101/api/command {"cmd": "COLLECT"}`

ThingSpeak Cloud API (slow path, history/alerts):

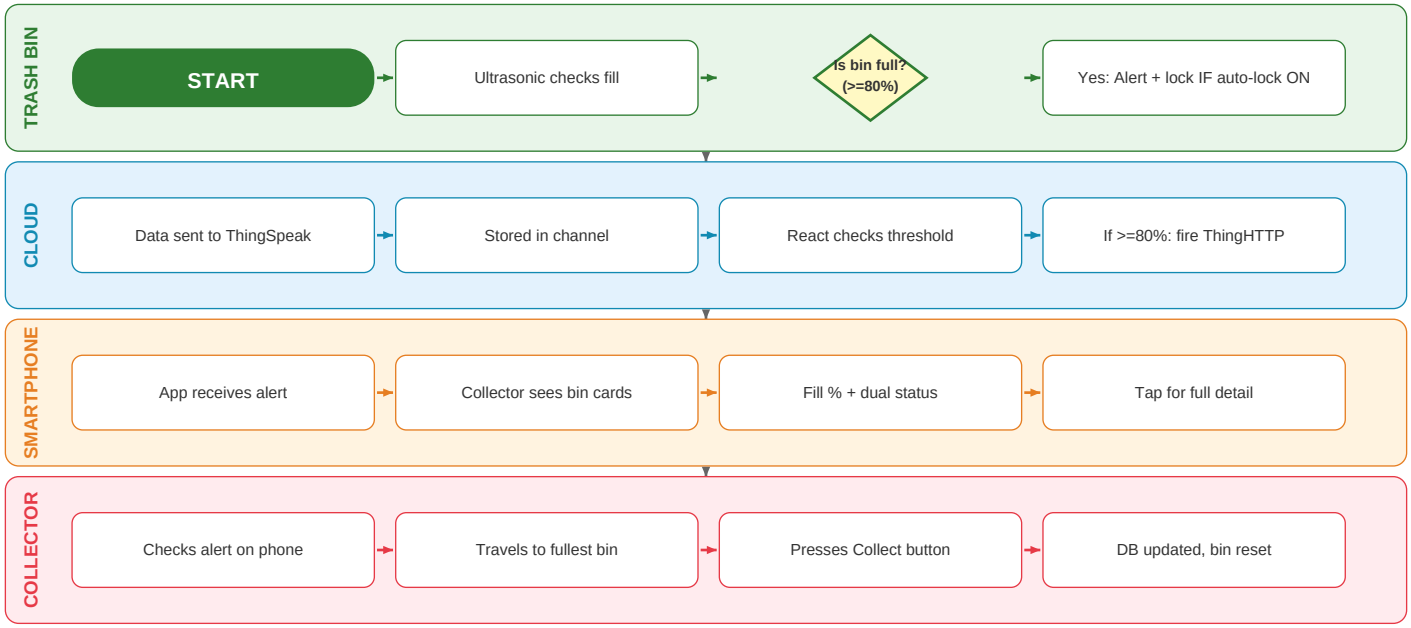
Read latest: `GET https://api.thingspeak.com/channels/{ID}/feeds/last.json?api_key={KEY}`

Read history: `GET https://api.thingspeak.com/channels/{ID}/feeds.json?results=50&api_key={KEY}`

Send command: `POST https://api.thingspeak.com/talkbacks/{TB_ID}/commands.json`

Complete Workflow (4 Lanes)

This is the swimlane flow from your presentation, with all corrections applied.



Manual LOCK Override (Separate Path)

The manual LOCK is independent from the automatic threshold flow. An admin/operator can press the LOCK button on the web dashboard at any time, regardless of fill level. This sends a LOCK command via local REST API (POST /api/command). The ESP32 processes it instantly and sets the LOCK flag. This is shown as a dashed line in the workflow diagram, bypassing the normal sensor-triggered path.

What Happens When Collector Presses 'Collect'

- 1. Collector taps COLLECT (app, dashboard, or touchscreen PIN)
- 2. Command via local API (instant) or TalkBack queue (cloud fallback)
- 3. ESP32 enters COLLECT mode: lid opens, PIR ignored, tamper skipped, buzzer silenced
- 4. Touchscreen shows COLLECTING with DONE button. Lid stays open.
- 5. Collector empties bin, taps DONE (no PIN needed) or operator sends COLLECT_DONE
- 6. ESP32 exits COLLECT mode: lid closes, all monitoring resumes
- 7. Fill recalibrates (~0%), LEDs go green, ThingSpeak updates
- 8. Dashboard and app reflect reset on next poll

Team Responsibilities

Brian — Schematic Design / PCB Layout

- Schematic done (KiCad, 3 sheets, BME680 version)
- PCB layout if feasible
- Docs: Ch. I 1.1 (with Banogbanog)
- Docs: Ch. II 8 Flowchart, 9 Schematic, 10 Components

Reyes — Firmware + Web Dashboard

- Write Arduino IDE code for ESP32 (sensor loop, WiFi, ThingSpeak API)
- Set up ThingSpeak channels, React, TalkBack, ThingHTTP
- Connect dashboard v10 (local .html) to live bin API
- Docs: 1.2, 3, 5 (w/ Banogbanog); Ch. II 7, 8

Pepito — Android Mobile App

- Build Android Studio project (Java or Kotlin)
- Implement home screen, bin detail, alerts, settings
- Push alerts (FCM or polling fallback), app closed
- Docs: Ch. I 2 RRL + 4 Framework (w/ Gunayan)

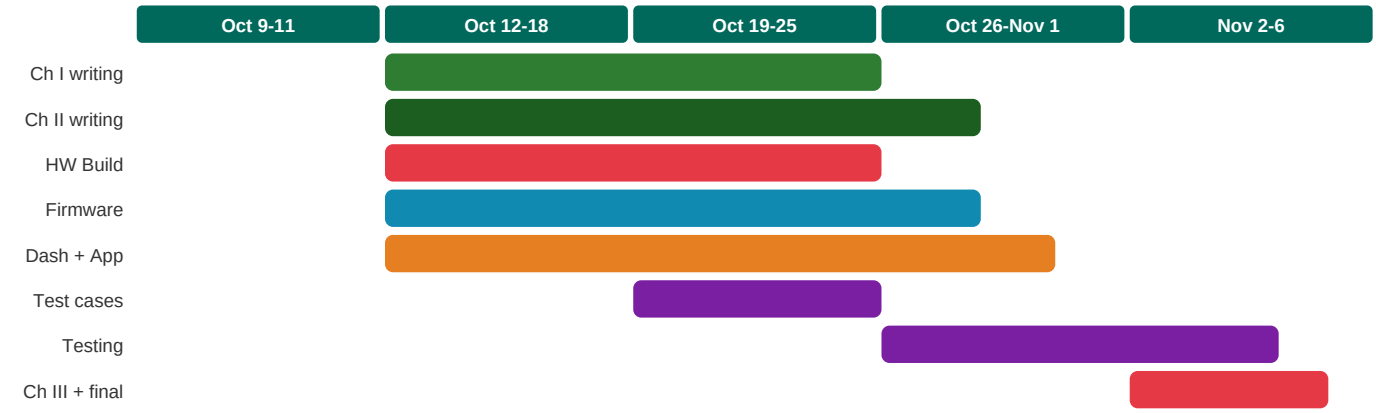
Banogbanog — Hardware Build / Physical Model

- Build table-size bin enclosures (2 bins)
- Wire components per Brian's schematic
- Divider wall + sealed power compartment
- Docs: 1.1, 1.2, 3, 5; Ch. II 6 Prototype

Gunayan — Hardware Assembly Assistant

- Assist Banogbanog with physical construction
- Help with sensor mounting and cable routing
- Test components; help integration testing
- Docs: Ch. I 2 RRL + 4 Framework (w/ Pepito)

5-Week Timeline (deadline Fri Nov 6)



Components List (x2 bins unless noted)

Component	Qty	Unit (PHP)	Subtotal	Shopee Store / Source
ESP32 DevKit V1 (38-pin WROOM-32D)	2	250	0 (owned)	Already owned
HC-SR04 Ultrasonic Sensor	2	45	0 (owned)	Already owned
HC-SR501 PIR Motion Sensor	2	55	0 (owned)	Already owned
SG90 Micro Servo Motor (180 deg)	2	65	0 (owned)	Already owned
RGB LED (Common Cathode, 5mm)	2	8	16	Shopee (pack of 10~PHP..
2.4" TFT Touch Display (ILI9341+XPT2046)	2	280	0 (owned)	Already owned
Passive Buzzer Module (5V)	2	12	24	Shopee (pack of 10~PHP..
BME680 Env. Sensor (replaces DHT22+MQ-135)	2	0	0	PRICE TBD: check Shop..
Resistors 1k + 2.2k (HC-SR04 echo divider)	2	0	0	In assorted pack below
HX711 Amplifier Board	2	60	120	Shopee (various sellers)
BMI160 6-Axis Inertial Sensor Module	2	80	0 (owned)	Already owned
5kg Straight Bar Load Cell	2	120	240	Shopee (various sellers)
4x18650 Battery Shield V9 (5V/3A out) -- Bin A	1	0	0 (owned)	Already owned
2x18650 Battery Shield (5V out) -- Bin B	1	0	0 (owned)	Already owned
Breadboard (830 tie-points)	2	75	150	Shopee / Lazada
Jumper Wires Kit (M-M, M-F, F-F)	2	55	110	Shopee (40pcs per set)
Micro USB Cable (data)	2	50	100	Shopee
Resistors Assorted (220, 1K, 10K)	1	50	50	Shopee (assorted pack)
Acrylic / Wood Enclosure Material	1	500	500	Hardware store / Shopee
Misc (screws, glue, tape, standoffs)	1	250	250	Hardware store

TOTAL TO BUY **PHP 1,560.00**

Gray rows already owned (ESP32 x2, HC-SR04 x2, PIR x2, SG90 x2, both battery shields, 2x TFT touch, 2x BMI160) -- PHP 1,550 market value not counted above.

Cost Split per Member (5 members)

Reyes	Pepito	Banogbanog	Gunayan	Brian
Firmware + Web Dashboard	Android App Developer	Hardware Build + Model	Hardware Assembly Assist	Schematic + PCB Layout
PHP 312	PHP 312	PHP 312	PHP 312	PHP 312

Notes

- Prices from the Sep 2026 Shopee check. BME680 price NOT included in the total: get a quote and add it (x2).
- DHT22/DHT11 and MQ-135 are no longer used (BME680 covers temperature, humidity, pressure and gas).
- Buying in bulk (e.g., 10-pack LEDs, buzzer packs) is cheaper than individual units shown above.
- Enclosure cost depends on material choice: acrylic sheet (~PHP 300-500), wood (~PHP 150-300), or 3D print (free if school has printer).
- USB cables and some resistors may already be available from previous labs -- check before buying.
- Consider ordering from one seller to combine shipping and save on delivery fees.
- CHECK: confirm the 2x18650 shield's rated output current. The 4x18650 (3A) easily covers ESP32+servo+sensors; a 2x18650 may only supply ~1-2A, which is tighter given the SG90 servo's stall current -- test Bin B under full load before relying on it.

Pin Assignment Table (per bin -- each bin has its own ESP32)

GPIO Pin	Component	Signal Type	Notes
GPIO 5	HC-SR04 Trig	Digital OUT	Trigger pulse (10us HIGH)
GPIO 12	Buzzer	PWM	Passive buzzer, frequency-controlled tones
GPIO 13	SG90 Servo	PWM	Lid open/close (50Hz, 1-2ms pulse)
GPIO 14	HX711 SCK	Digital OUT	Load cell clock line
GPIO 15	TFT Touch CS	SPI (shared bus)	Touch chip-select, idle HIGH (matches boot strap d..
GPIO 16	TFT D/C	Digital OUT	Data/command select for display
GPIO 17	TFT CS	Digital OUT	Display chip-select
GPIO 18	HC-SR04 Echo	Digital IN	Via 1k/2.2k divider: 5V echo -> ~3.4V safe for ESP32
GPIO 19	TFT MOSI	SPI (shared bus)	Also wired to Touch T_DIN pad
GPIO 21	I2C SDA	I2C Data	BMI160 (0x68, SA0=GND) + BME680 (0x76, SDO=..
GPIO 22	I2C SCL	I2C Clock	BMI160 (0x68) + BME680 (0x76); both CS pins tied..
GPIO 23	TFT SCK	SPI (shared bus)	Also wired to Touch T_CLK pad
GPIO 25	RGB LED Red	Digital OUT	Red channel (active HIGH)
GPIO 26	RGB LED Green	Digital OUT	Green channel (active HIGH)
GPIO 27	PIR Sensor	Digital IN (IRQ)	Interrupt-based motion detect
GPIO 32	HX711 DT	Digital IN	Load cell data line
GPIO 33	RGB LED Blue	Digital OUT	Blue channel (active HIGH)
GPIO 35	TFT MISO	SPI IN (input-only)	Also wired to Touch T_DO pad
GPIO 36	TFT Touch IRQ	Digital IN (input-only)	Optional: signals when screen is touched

I2C Bus Sharing

I2C BUS (GPIO 21 SDA, GPIO 22 SCL) -- BMI160 + BME680

- BMI160 at address 0x68 (SA0=GND) -- accelerometer/gyro for tamper detection
- BME680 at address 0x76 (SDO=GND, CS=3.3V) -- temperature, humidity, pressure, gas (IAQ)
- TFT + touch run on SPI instead (GPIO 15,16,17,19,23,35,36) -- see table above

Pin Usage Summary

19 PINS USED

GPIO 5,12,13,14,15,16,17,18,19,
21,22,23,25,26,27,32,33,35,36

6 sensors + 4 actuators + shared I2C + TFT/touch SPI

FREE PINS

GPIO 4, 34, 39 (+ GPIO 2 strapping)
(GPIO 0, EN not usable as I/O)

BME680 swap freed GPIO 4 and 34

NOTE: GPIO 34-39 are input-only (no pull-up). TFT MISO uses 35, Touch IRQ uses 36. TFT VCC + LED backlight on 5V, RESET to 3.3V. GPIO 0, 2, 12, 15 are strapping pins. GPIO 15 (Touch CS) defaults HIGH at boot, which matches its idle state, so it's safe here.

Why This Matters for the Demo/Defense

Sensor values and calibration constants used to be hardcoded in firmware, meaning every demo tweak required reflashing the ESP32. Now both are adjustable live from the dashboard, with zero code changes.

Per-Sensor Simulation Control (Auto / Manual)

Sensor	Auto Mode	Manual Mode	Manual Input
Fill Level (HC-SR04)	Reads real distance	Operator sets %	Slider 0-100%
Weight (HX711)	Reads load cell	Operator sets kg	Number input, 0-5 kg
Air quality (BME680)	Reads IAQ index	Operator sets IAQ	Stepper 0-500
Temp/Humidity (BME680)	Reads real values	Operator sets both	Two steppers
Tamper (BMI160)	Reads tilt angle	Operator force-triggers	Toggle button
Motion (PIR)	Reads interrupt pin	Operator force-triggers	Toggle button
Lid (SG90 Servo)	Firmware auto-controls	Operator direct control	Open / Close buttons

MIXED STATES OK: Each sensor toggles independently. Fill can be Manual while Gas stays Auto -- demo one failure at a time. Applies per bin, so Bin A and Bin B can run different simulation states simultaneously.

Role Access

OPERATOR (Web Dashboard)

- Full control: toggle Auto/Manual per sensor
- Sets manual values for any sensor
- Edits all Calibration Settings (Channel 3)
- Sees SIMULATION MODE banner when active

COLLECTOR (Mobile App)

- Read-only: sees current values, real or simulated
- Cannot toggle Auto/Manual or edit calibration
- Sees SIMULATION MODE banner when active
- Focused on pickup tasks (Collect, route, alerts)

User-Configurable Calibration Settings (Channel 3)

Setting	Default	What It Controls
Distance Empty	**30 cm**	HC-SR04 reading when bin is completely empty (calibration reference)
Distance Full	**5 cm**	HC-SR04 reading when bin is considered full (calibration reference)
Fill Threshold	**80%**	Fill level that triggers alert, LED red, and optional auto-lock
Temp Threshold	**40 C**	Temperature spike limit that triggers alert
Gas Threshold	**200 IAQ**	BME680 air-quality index treated as odor alert (range 50-500)
Tilt Threshold	**30 deg**	BMI160 tilt angle that triggers tamper alarm
Weight Capacity	**4500 g**	Load reading considered near-full, backs up fill % reading
Auto-Lock Enabled	**OFF**	Whether the bin auto-locks its lid once Fill Threshold is hit
Servo Closed*	**0 deg**	Servo angle for lid closed position (stored in EEPROM, not ThingSpeak)
Servo Open*	**90 deg**	Servo angle for lid open position (stored in EEPROM, not ThingSpeak)
Collect PIN*	**1234**	4-digit PIN for touchscreen COLLECT (EEPROM; visible read-only on app)

NO RECODE NEEDED: All 11 settings above are editable from the dashboard via local API (POST /api/calibrate) and take effect instantly on the ESP32 -- no Arduino IDE, no reflashing, no recompiling required.

COLLECT MODE (Anti-Tamper Bypass):

COLLECT command received (local API, TalkBack, or touchscreen PIN) sets collectMode = true on ESP32:

- Touchscreen COLLECT requires 4-digit PIN (set in calibration, visible read-only on app)
- LOCK flag clears, lid opens and stays open (no auto-close, no auto-lock, PIR ignored)
- Tamper check (step 10) skipped -- collector can lift/tilt bin without triggering alarm
- Buzzer silenced -- no tamper tone, no threshold tone during active collection
- Touchscreen shows COLLECTING... status with a big DONE button (DONE needs no PIN)

Resume: Collector taps DONE on touchscreen or operator sends COLLECT_DONE via dashboard. All monitoring resumes.

TAMPER RECOVERY (Hybrid Auto-Clear):

BMI160 tilt > threshold triggers alarm tone (2kHz/500Hz alternating) and starts recovery countdown:

- Touchscreen shows TAMPER DETECTED with live countdown (10...9...8...)
- If tilt returns to stable within 10s: auto-clear, alarm stops, event logged to ThingSpeak
- If still unstable at 15s: flag locks, alarm continues, requires manual clear (dashboard only)
- Manual clear: operator presses CLEAR TAMPER on dashboard (operator-only control)

Design Direction: Dual-Style Approach

The Smart Bin system uses two complementary UI styles optimized for their target platform:

ESP32 Touchscreen = Neo-Brutalism (thick 3px borders, hard 4px shadows, flat fills, 4px radius) for max TFT readability.

Dashboard + Mobile App = Brutal-Morph (Neo-Brutalism structure + Neumorphism depth) for refined web/mobile UIs.

Style Comparison

NEO-BRUTALIST

ESP32 Touchscreen (240x320 TFT)

Borders:	3px solid, high-contrast
Shadows:	4px hard offset, no blur
Corner Radius:	4px (slightly rounded)
Typography:	Inter Bold / Regular
Cards:	Flat fill + thick border
Buttons:	Solid fill + shadow shift on press
Inputs:	Thick-bordered rectangles
Danger:	Red flat fill, bold text
Key Trait:	Raw, functional, maximum contrast

BRUTAL-MORPH

Web Dashboard + Mobile App

Borders:	2px solid (thinner, refined)
Shadows:	Dual neumorphic 6-8px blur + border
Corner Radius:	8px (softer than Brutalist)
Typography:	Inter Semi-Bold / Regular
Cards:	Raised neumorphic surface + 2px border
Buttons:	Neumorphic press effect + 2px border
Inputs:	Inset neumorphic wells + border
Danger:	Flat Brutalist (hard, no neumorphism)
Key Trait:	Structure of Brutalism + depth of Neumorphism

Color Palette: Warm Concrete (Shared)

LIGHT MODE

Background	`#F5F0EB`	Warm off-white
Surface	`#FFFFFF`	Cards, panels
BM Background	`#E8E0D8`	Brutal-Morph base
Border/Shadow	`#1A1A1A`	Borders, hard shadows
Text Secondary	`#6B635B`	Labels, captions

DARK MODE

Background	`#2A2520`	Deep warm charcoal
Surface	`#3A3530`	Cards, panels
BM Background	`#2E2924`	Brutal-Morph dark base
Border	`#E8E0D8`	Inverted borders
Text Secondary	`#A89E94`	Labels, captions

Status Colors (Same Across All Interfaces & Modes)

Green	Amber	Red	Purple	Teal
OK / < 50%	50 - 79%	>= 80%	Simulation	Calibration

VISUAL CONSISTENCY: Both styles share the same Warm Concrete palette, status colors, and information hierarchy.
Danger elements always use flat Brutalist styling (hard borders, no neumorphism) in both Neo-Brutalist and Brutal-Morph.

All screens rendered via TFT_eSPI library on ILI9341 (portrait 240x320). Uses fillRect/drawRect for borders and shadows. Touch input via XPT2046. Inter font approximated with built-in fonts. Warm Concrete palette throughout.

Screen Inventory (8 States)

A1: Main Status (Default Idle)

When: Normal operation, no alerts

- Fill % large centered (bold, 48pt equivalent)
- Vertical fill bar on left edge, color-coded by level
- Lid Status label: OPEN / CLOSED (top-right)
- Bin Status label: OK / LOCKED / TAMPER / COLLECTING
- Background tinted by status color (green/amber/red)

A2: Alert Screen (>= 80% Full)

When: Fill level reaches 80% threshold

- Red (#D93025) background, high urgency
- Large URGENT header in bold white text
- Fill percentage displayed prominently
- Buzzer-active indicator icon
- Auto-transitions from main status screen

A3: COLLECT Flow (PIN Entry)

When: Operator/collector initiates collection

- Enter PIN header text at top
- 4-digit PIN display (masked with dots)
- 3x4 numeric keypad grid (chunky 3px-border buttons)
- CANCEL button (bottom-left, red accent)
- Wrong PIN = flash red border, allow retry

A4: COLLECTING State

When: Correct PIN entered, servo opens lid

- COLLECTING... large text centered
- Servo status indicator: LID OPEN
- Green accent background
- Lid Status = OPEN, Bin Status = COLLECTING
- Waiting for collector to finish

A5: DONE Screen

When: Collection complete, ready to close

- COLLECTION COMPLETE header text
- Large DONE button (green, full-width, 3px border)
- No PIN needed (lid already open)
- Auto-returns to main status after 30s timeout
- Servo closes lid when DONE pressed

A6: TAMPER Alert

When: BMI160 IMU detects abnormal movement

- Red flashing border effect (alternating rect strokes)
- TAMPER DETECTED bold centered text
- Live countdown timer (10s normal / 15s locked)
- Buzzer active indicator (PWM on GPIO 12)
- Returns to normal when countdown expires

A7: LOCKED Screen

When: Operator sets software lock via dashboard

- LOCKED large text in gray/purple
- Lock icon (drawn with rectangles)
- Contact Operator message below
- No action buttons available to user
- Software lock flag -- not a servo position

A8: Startup / Calibration

When: System boot or manual calibration

- Teal (#0D8A8A) themed background
- CALIBRATING... text with progress indicators
- Per-sensor init status (HC-SR04, BME680, HX711, BMI160)
- System info: IP address, firmware version
- Transitions to Main Status when complete

IMPLEMENTATION: All screens use TFT_eSPI drawRect (3px) for borders, fillRect offset by 4px for hard shadows, and fillScreen with Warm Concrete hex values. Touch zones mapped to button coordinates via XPT2046 readings.

Brutal-Morph = Neo-Brutalism structure (2px borders, flat headers, chunky type) + Neumorphism surfaces (dual light/dark shadows, 6-8px blur, raised/inset panels). Danger elements stay flat and hard (Brutalist).

Brutal-Morph Shadows (Light): Light: #F5F0EB (top-left) | Dark: #C4BAB0 (bottom-right) | Base: #E8E0D8 | 8px radius, 2px borders

D Dashboard Views (5 Screens)

B1: Main Dashboard (Bin Overview)

When: Default view after page load

- Side-by-side bin cards (raised neumorphic + 2px border)
- Per-bin: fill %, fill bar, status color strip (flat Brutalist)
- Sensor readings grid: temp, humidity, gas, weight
- LOCK / UNLOCK toggle (neumorphic press + 2px border)
- Connection indicator: Local REST (green) / ThingSpeak (blue)
- Simulation mode badge: solid purple (flat Brutalist)

B2: Settings Panel

When: Operator clicks Settings tab

- Fill threshold sliders (warning % / critical %)
- WiFi / AP configuration inputs (inset neumorphic wells)
- Data refresh interval selector (dropdown, 2px border)
- Buzzer and LED alert enable/disable toggles
- TAMPER sensitivity slider (BMI160 IMU threshold)
- Save / Reset buttons (raised neumorphic + 2px border)

B3: Simulation Controls

When: Operator enables simulation mode

- Solid purple SIMULATION banner across top (flat Brutalist)
- Per-sensor toggle switches (on/off per reading)
- Manual value input fields (inset neumorphic wells)
- Simulated vs Real indicator badge per sensor
- Apply Changes / Reset to Real buttons
- Warning callout: Values are simulated (flat orange border)

B4: Charts & History

When: Operator clicks Charts tab

- ThingSpeak-embedded chart iframes (2px bordered containers)
- Fill level over time (line graph, green/amber/red zones)
- Temperature + humidity dual-axis chart
- Environment chart: ALL / temp / humidity / gas (IAQ) / pressure
- Time range selector: 1h / 6h / 24h / 7d buttons
- Refresh button + auto-refresh interval display

B5: Collector Management

When: Operator clicks Collectors tab

- Collector account list (neumorphic card per collector)
- 4-digit PIN assignment per collector (editable)
- Collection history log (timestamped entries)
- Add Collector / Remove buttons (neumorphic + 2px border)
- Active collection status: which bin, which collector

HYBRID RULE: Informational surfaces (bin cards, sensor panels, charts) use raised neumorphic styling with soft shadows. Danger/alert elements (URGENT banner, LOCK badge, simulation warning) stay flat Brutalist -- no neumorphism, hard borders only.

Same Brutal-Morph principles as dashboard: neumorphic surfaces + 2px borders for info, flat Brutalist for danger. Vertical layout, larger touch targets (min 48dp), bottom tab navigation with thick 2px top border.

M App Screens (8 States)

C1: Home Screen (Bin List)

When: App opens, default view

- Stacked bin cards (vertical, not side-by-side)
- Each card: raised neumorphic surface + 2px border
- Fill % + status color strip per bin (flat Brutalist)
- Pull-to-refresh gesture for data update
- Bottom tab bar with thick 2px top border

C2: Simulation Banner

When: System is in simulation mode

- Sticky purple banner pinned to top of screen
- SIMULATION MODE bold white text
- Lists which sensor values are simulated
- Flat Brutalist styling (danger/warning element)
- Tap banner to view simulation details

C3: Bin Detail View

When: User taps a bin card from home

- Expanded single-bin view (full screen)
- All sensor readings: temp, humidity, gas, weight
- Large fill bar + percentage (neumorphic container)
- COLLECT button (large, neumorphic press + 2px border)
- Bin ID, location info, last-updated timestamp

C4: Urgent Alert

When: Fill >= 80% push notification

- Full-screen red overlay (flat Brutalist, no neumorphism)
- URGENT: Bin Nearly Full header in bold white
- Fill % and which bin (Bin 1 or Bin 2)
- Collect Now CTA button (flat red, 2px white border)
- Dismissible after user acknowledges

C5: Locked Bin View

When: Viewing a bin the operator has locked

- Grayed/muted bin card (reduced opacity surface)
- LOCKED badge (flat Brutalist, 2px border)
- Contact Operator message displayed below
- COLLECT button hidden (not just disabled)
- Shows last-known sensor readings (stale indicator)

C6: Alert History

When: User opens Alerts tab

- Scrollable vertical list of past alerts
- Each alert: timestamp + type + bin ID
- Color-coded severity stripe (green/amber/red)
- Hard shadow on alert cards (Brutalist element)
- Filter by bin / severity controls at top

C7: Settings

When: User opens Settings tab

- Server URL / IP config (inset neumorphic input wells)
- Notification preference toggles
- Theme toggle: Light / Dark mode switch
- About section: app version, group info
- Logout button (flat Brutalist, red accent)

C8: Push Notification

When: Alert triggers Android notification

- Android notification card in system tray
- Alert type icon + severity color accent
- Bin ID + fill level in notification body
- Tap-to-open navigates to bin detail (C3)
- Brutalist-styled app icon badge

COLLECTOR ROLE: The mobile app is read-only except for the COLLECT action. Collectors cannot LOCK/UNLOCK bins, change thresholds, or toggle simulation. Those controls are operator-only via the web dashboard (see page 19).

TFT LCD Orientation on Bin

RECOMMENDED: Portrait (vertical)

The front face of a bin is taller than wide -- portrait fills that space naturally.
Fill level displays as a vertical bar matching the actual bin contents rising upward.
Status text (bin name, weight, gas level) stacks cleanly in portrait layout.

PORTRAIT (recommended)

- Vertical fill bar matches bin shape
- Clean text stacking (name, weight, %)
- Fits taller-than-wide bin front panel
- Better for at-a-glance status reading
- 240 x 320 px (W x H) in portrait

LANDSCAPE (alternative)

- Wider layout for charts/graphs
- Better if showing bar charts side-by-side
- 320 x 240 px (W x H) in landscape
- Not ideal for a typical bin front face
- Only use if UI demands wide layout

BMI160 Sensor Mounting & Accuracy

Accuracy is NOT affected by mounting orientation

The BMI160 measures acceleration on all 3 axes (X, Y, Z) regardless of board orientation.

What matters:

1. Firmware calibration -- on startup, read resting orientation as baseline reference vector
2. Rigid mounting -- BMI160 must be firmly attached to PCB and PCB firmly to the bin body

TIP: If BMI160 is flat (Z-axis up), compare against gravity on Z. If mounted sideways, same math, different axis. Set the reference vector during firmware calibration. Tilt detection math adjusts automatically.

Component Priority (If Pins Run Short)

Priority	Component	Pins	Why / Notes
LOW	**RGB LED**	3	Redundant if TFT shows status. Replace with 1 single LED to save 2 pins.
LOW	**TFT Touch IRQ (GPIO 36)**	1	Optional -- can poll touch instead of interrupt. Easy 1-pin save.
LOW	**TFT Touchscreen (full)**	7	Nuclear option. Replace with I2C OLED (shares BMI160 bus, 0 extra pins).
MED	**Buzzer**	1	Alerts are nice but not critical. Rely on LED/TFT for warnings.
MED	**PIR Sensor**	1	Can detect approach via ultrasonic distance drop instead.
MED	**SG90 Servo**	1	Auto-open lid is convenience only. Manual lid still works.
HIGH	**HC-SR04 Ultrasonic**	2	Core: fill level is the primary measurement of the bin.
HIGH	**HX711 + Load Cell**	2	Core: weight measurement is primary function.
HIGH	**BME680**	0*	Temp, humidity, pressure and gas in one module. *Shares I2C bus.
HIGH	**BMI160**	2*	Anti-tamper security. *Uses shared I2C bus, no extra pin cost.

QUICK WIN: Remove RGB LED (save 3 pins) + drop Touch IRQ (save 1 pin) = 4 pins freed with zero functional loss. The TFT already shows status colors, and touch polling works fine without the IRQ interrupt line.

ESTIMATED POWER CONSUMPTION

Per Bin Unit -- ESP32 + All Peripherals

All values are typical operating current at 3.3V logic / 5V supply unless noted. Actual draw varies with activity (WiFi TX bursts, servo movement, buzzer pulses). Each bin unit has its own ESP32 and full sensor set. Two bins = double the total system draw. Current values from component datasheets and empirical measurements.

A Component Current Draw

Component	Active (mA)	Idle (mA)	Notes
ESP32 DevKit V1	160-240	20	WiFi active ~200mA avg; modem-sleep ~20mA; deep-sleep ~0.01mA
HC-SR04 Ultrasonic	15	2	5 readings per cycle (median filter); idle between cycles
BME680 (I2C)	up to 12	<0.01	Short gas-heater pulses per reading; sleeps between reads (datasheet)
HX711 + Load Cell	1.5	0.001	Can power-down between reads (<1uA); active during weigh
BMI160 IMU (I2C)	0.9	0.003	Low-power accel mode available; shared I2C bus with TFT touch
SG90 Servo	100-250	10	Peak during movement; idle = holding torque (~10mA)
PIR (HC-SR501)	0.065	0.065	Always-on, interrupt-driven; essentially negligible
ILI9341 TFT 2.4"	80-100	80	Backlight dominates; ~20mA without backlight
RGB LED (1 channel)	20	0	20mA per color; 60mA max all 3 on; 0 when off
Passive Buzzer	30	0	PWM-driven on GPIO 12; 0 when silent

E Operating Scenarios (Per Bin)

Typical Active **~345 mA**

WiFi on, all sensors reading, TFT backlight on, LED green, servo idle, buzzer off

Peak Draw **~600 mA**

Servo moving + buzzer alarm + all LEDs + WiFi TX burst (worst-case simultaneous)

Idle (No Motion) **~310 mA**

Same as typical but PIR quiet, servo idle -- ESP32 Wi-Fi and TFT backlight dominate

Demo / Defense **~345 mA**

Powered by USB (500mA limit) -- may need powered hub or wall adapter for servo peaks

B Battery Runtime Estimate

Assume 18650 cells of 3.7V / 2600 mAh, boosted to 5V at ~85% efficiency: 2600 x 3.7 x 0.85 / 5 = ~1,635 mAh per cell.

Bin A (4x18650 shield): ~6,540 mAh / ~345 mA typical = ~19 hours
Bin B (2x18650 shield): ~3,270 mAh / ~345 mA typical = ~9.5 hours

Removing the MQ-135 (~150 mA heater) cut typical draw from ~490 mA to ~345 mA. Estimates only: measure real current with a USB meter during testing and record it in Chapter III.

P Recommended Power Supply

For Demo / Defense

USB-C or Micro-USB wall adapter, 5V 2A minimum (covers peak with margin)
USB from laptop may hit 500mA limit -- use powered USB hub if servo stutters
Both bins can share one 5V 2A+ adapter with a USB splitter

For Deployment

18650 battery holder + TP4056 charger module + 5V boost converter
Consider solar panel (6V 1W) + charge controller for outdoor unattended use
Lower TFT backlight or Wi-Fi modem-sleep if battery life is a priority

BME680 SWAP SAVED ~30%: Dropping the always-on MQ-135 heater took typical draw from ~490mA to ~345mA per bin. The biggest remaining loads are ESP32 Wi-Fi (~200 mA) and the TFT backlight (~80 mA).

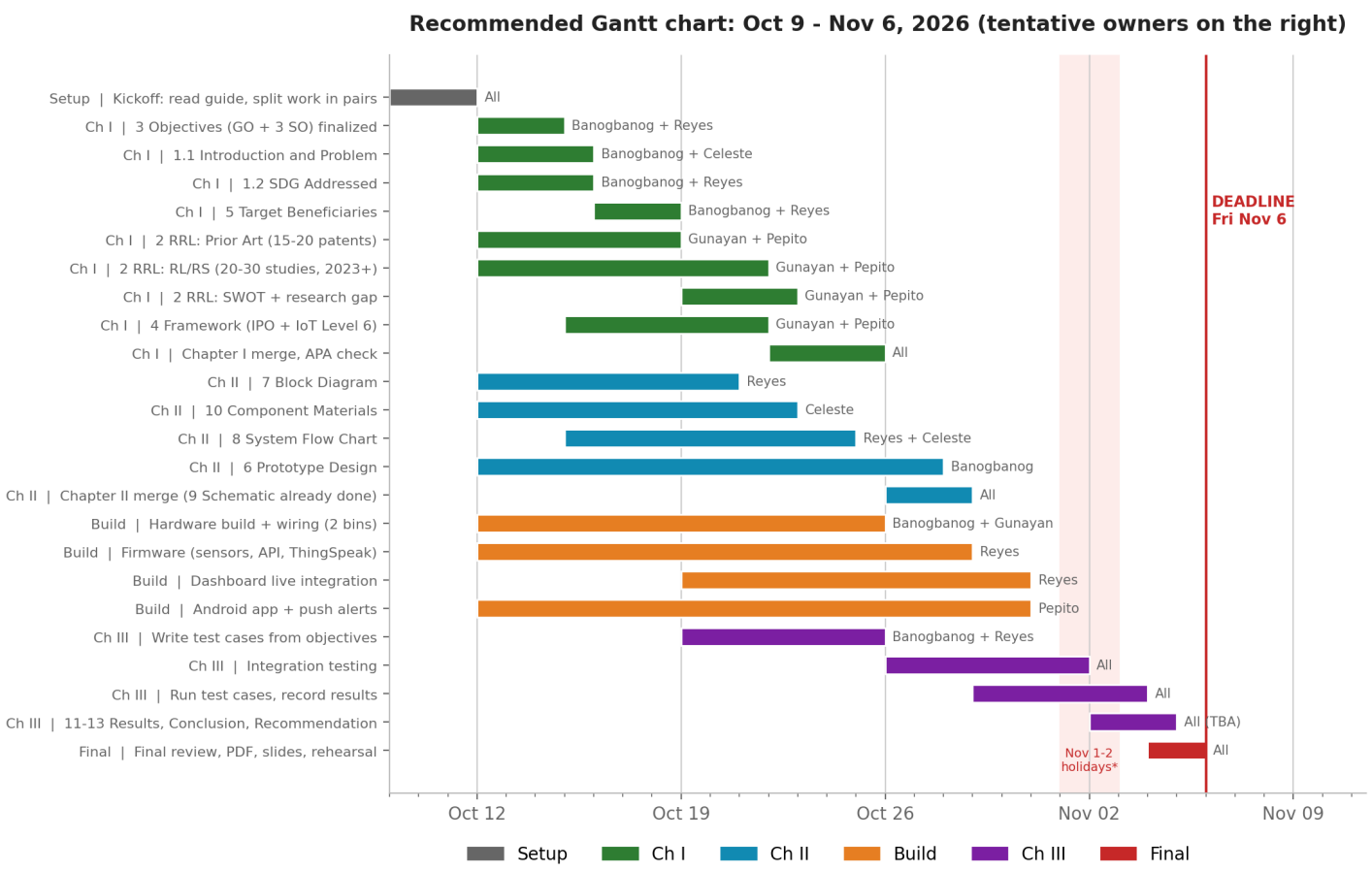
Chapter Assignments (tentative, may change)

Ch I 1.1	Introduction & Problem (200-300 words)	**Banogbanog, Celeste**
Ch I 1.2	SDG Addressed (1 SDG: SDG 11, target 11.6)	**Banogbanog, Reyes**
Ch I 2	Review of Literature: Prior Art 15-20, RL/RS 20-30 (2023+), SWOT + research gap	**Gunayan, Pepito**
Ch I 3	Objectives: General = SO1 + SO2 + SO3 (each one testable)	**Banogbanog, Reyes**
Ch I 4	Framework (IPO + IoT Level 6, from the Miro diagram)	**Gunayan, Pepito**
Ch I 5	Target Beneficiaries	**Banogbanog, Reyes**
Ch II 6 / 7	Prototype Design / Block Diagram	**Banogbanog / Reyes**
Ch II 8	System Flow Chart	**Reyes, Celeste**
Ch II 9 / 10	Schematic Design (done) / Component Materials	**Celeste**
Ch III 11-13	Test cases per objective, results, conclusion, recommendation	**All (to be assigned)**

Objectives -> Test Cases (prof: Chapter III tests each objective)

GO	Design and develop an ESP32-based IoT Smart-Bin system that monitors fill level, weight and bin air, detects tampering, and gives operators and collectors real-time status, alerts and collection priority through a web dashboard, Android app and on-bin touchscreen.	**all TCs**
SO1	Build two bin units that measure fill (HC-SR04), weight (HX711), temperature/humidity/pressure/IAQ (BME680) and tilt (BMI160) within set accuracy limits, with PIR auto-lid and tamper alarm.	**TC1.1-TC1.6**
SO2	Develop the web dashboard and Android app to show live readings from both bins within 2 s, log history to ThingSpeak every 15 s, and deliver alerts (banner + push) within 20 s of a threshold or tamper event.	**TC2.1-TC2.6**
SO3	Implement operator and collector functions: collection priority + time-to-full, remote LOCK/UNLOCK, PIN-protected COLLECT/DONE, per-sensor simulation and calibration applied without reflashing.	**TC3.1-TC3.6**

Recommended Gantt Chart (Oct 9 - Nov 6, 2026)



*Nov 1 All Saints' Day (Sunday); Nov 2 is usually declared a special non-working day. Check the TUP calendar.